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ASSIGNMENT 1

Algoritmos de Diagnóstico e de Auto-Regulação

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1. Introduction

The general objective of this work is to analyse the motor activities of a group of volunteers using the signals provided by a smartphone. The model of the smartphone used for data collection was *Samsung Galaxy S II* and it was placed on the waistline during the experiments.

The data analysed were the 3-axial linear accelerations and 3-axial angular velocities acquired using the embedded accelerometers and gyroscopes. The signals were sampled at 50 Hz. In total 12 activities were analysed, divided into three categories as follows:

1. Dynamic
    - Walking
    - Walking up
    - Walking down
2. Static
    - Sitting
    - Standing
    - Lying
3. Transition
    - Sitting down
    - Standing up
    - Lying down
    - Sitting up
    - Lying down from standing
    - Standing up from lying

Tracking human movements has many purposes for our lives. A good example is the development of *Active Assisted Living* (AAL) solutions, which focus on the elderly population. These applications monitor abnormal movements and can detect incidents, such as falls, heart attacks, etc. With the help of an effective alarm system, it is possible to intervene instantly by calling an ambulance.

It is possible to quantify the quality of the results as a function of the accuracy with which the classification of the activities can be made. Firstly, separating the more general categories (dynamic, static, transition) and then the subcategories.

The quality of the clustering, evaluated in terms of accuracy and silhouette, depends on the settings and on the application of various signal processing procedures, as summarised in this diagram:

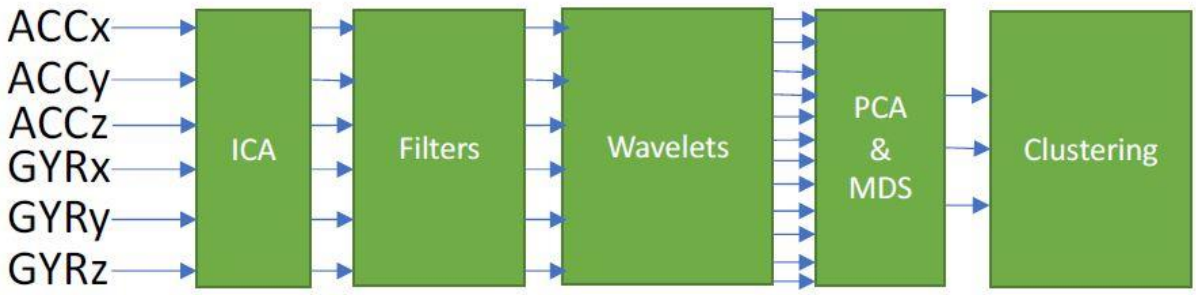


***Figure 1**: Multi-step methodology*

In the following sections, the principles of operation will be described and the effects of each method mentioned above will be analysed. This will be done through the analysis of the results obtained after implementation in MATLAB.

2. Data collection and analysis

We were provided with 11 ‘data.mat’ packages, with each data file representing an experiment performed by one volunteer (usually two experiments per volunteer were performed) and containing this information:

- **Fs (1x1)**: Sampling frequency.
- **Sensors (1x6)**: Cell array containing the names of the sensors.
- **activities (1x12)**: Cell array containing the activity labels.
- **data (Nx6)**: Matrix containing the time series for the six sensors, being N the number of samples.
- **label_data (Ax5)**: Matrix containing annotations, where each line corresponds to one activity, and A is the total number of activities labelled for a given file. The first two columns correspond to the experiment and user IDs, as indicated in the file name. The fourth and fifth columns indicate the start and the end samples of a given activity. The third column corresponds to the activity number and can be used to index the cell array “activities” to obtain the related activity name.
- **t (1xN)**: Time vector related to the time series.

In the example below six raw signals are reported - one for each sensor - extracted from the ‘data.mat’ of a single user. Different colours were used to annotate different activities. Black segments represent non-annotated parts of the signal.

Knowing that the possible activities are twelve, it can be noticed that some activities were repeated more than once during the single experience.

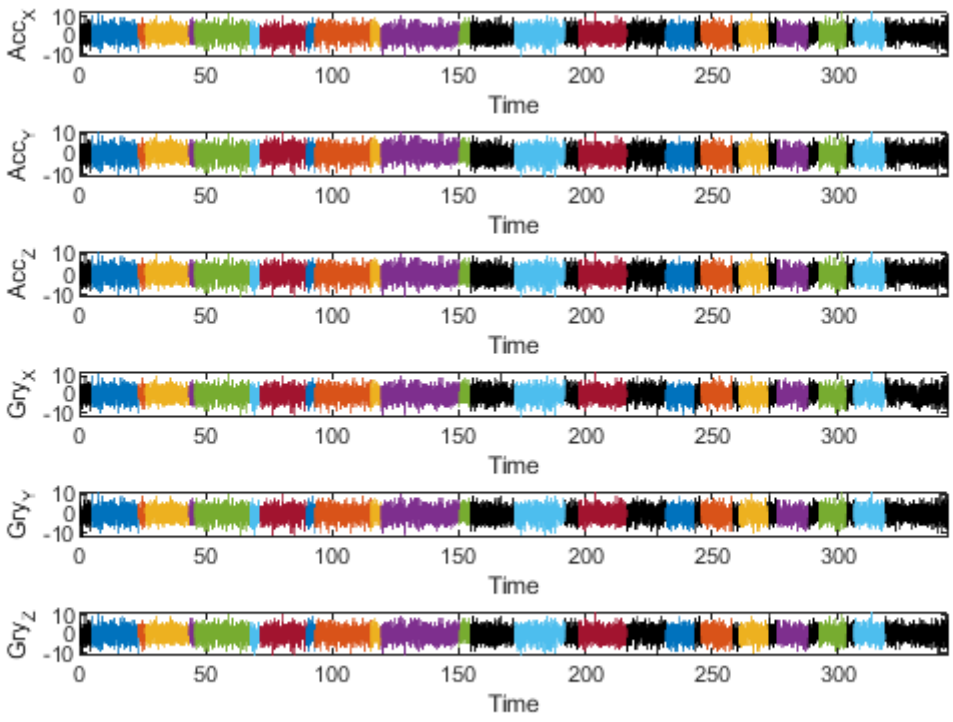


***Figure 2**: raw signals, (exp30_user15)*

3. Independent Component Analysis (ICA)

The measured signals contained noise that needed to be minimised. We applied different approaches to address the problem.

The first approach is to carry out the Independent Component Analysis, whose purpose is to separate n measured signals into n-independent components, assuming they have a non-Gaussian distribution and they are statistically independent. In other words, the ICA finds the independent components (also called the factors, latent variables or sources) by maximising the statistical independence of the estimated components. The relationship between the source ‘s’ and measured signal ‘x’ is quantified in the elements of matrix ‘A’, as explained below:

$$x = As \quad x = \sum_{i=1}^{n} a_i s_i \qquad x_j = a_{j1}s_1 + a_{j2}s_2 + \ldots + a_{ji}s_i + \ldots + a_{jn}s_n \quad \text{for all } j$$

The goal is to calculate the matrix that maximises this independence.[1] Several algorithms can be used. We used the fastICA algorithm, which consists in an iterative process: starting from an initial solution, the algorithm iterates to reach better solutions until the function eventually reaches the optimum. After applying the 'fastica' function [2] on the matrix ‘data’ containing the time series for the six sensors, it was observed that the noise is associated with the sixth source signal (but could be in another source signal), which is linearly dependent on the sixth[1] column of the 'A' matrix. For this reason, the elements in this column were set up to zero and the signals were reconstructed without the noise component and plotted again.

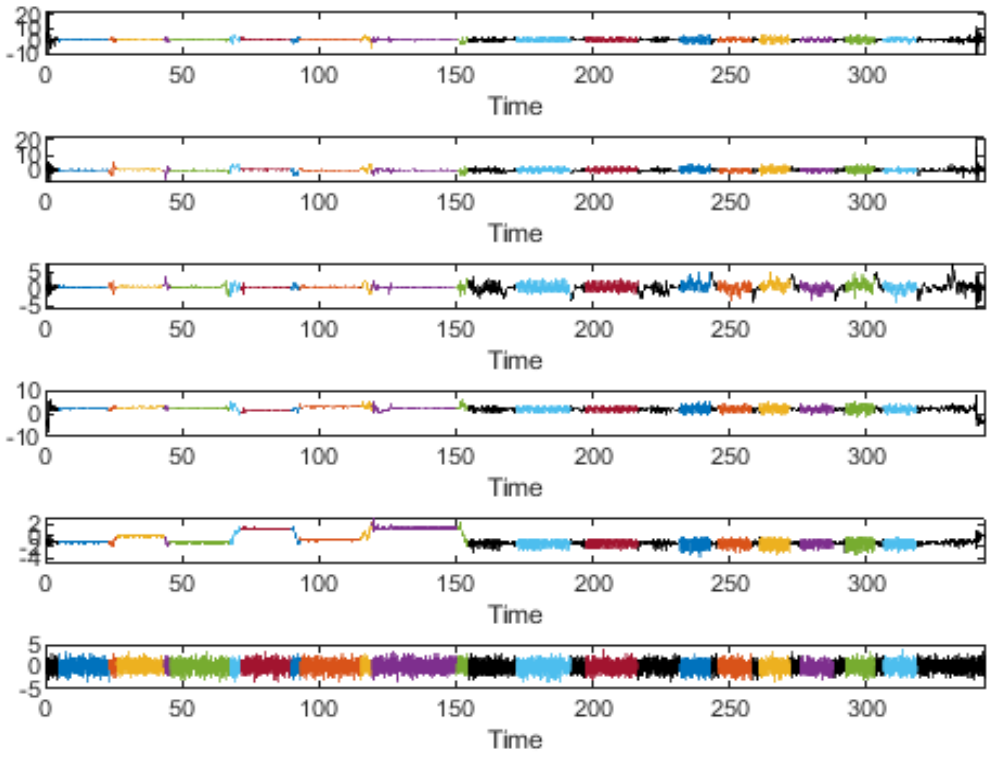


***Figure 3**: 'fast' matrix (sources) obtained by the 'fastica' function, (exp30_user15)*

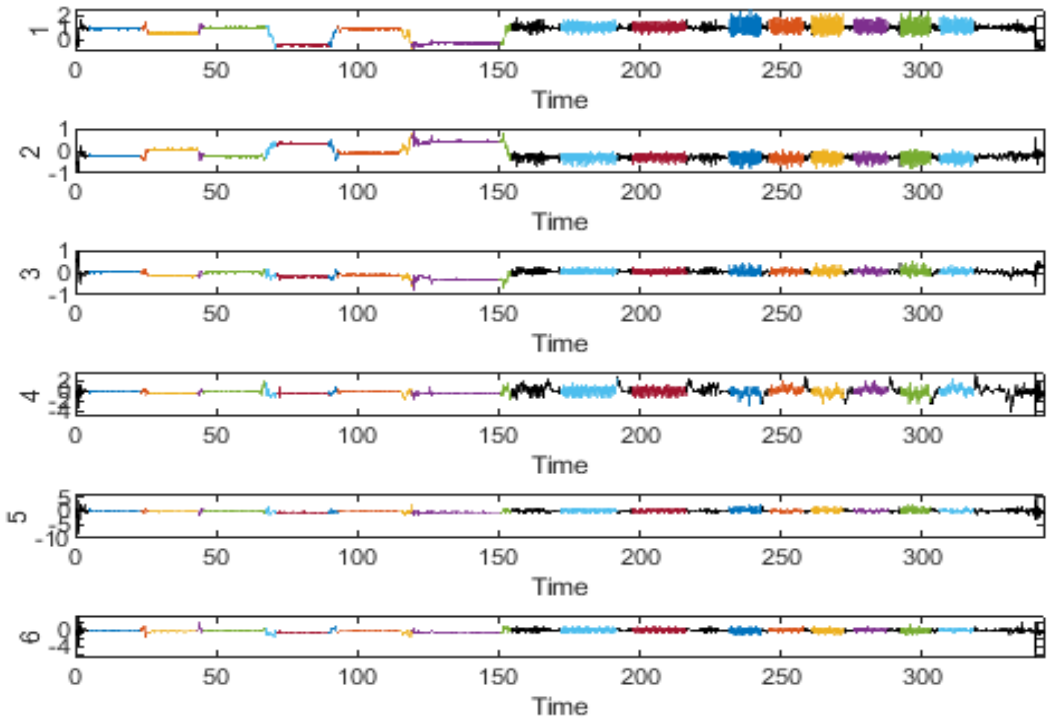


***Figure 4**: 'fast' matrix (sources) with A(:,6)=0 , (exp30_user15)*

4. Frequency filter

After obtaining the signals from the ICA, we proceeded to the filtering of the signal. Since the bandwidth of interest is [0-20] Hz we applied a filter that eliminates the frequencies outside the bandwidth. For that we used a low-pass FIR filter with a cut frequency of 20 Hz with the window type Hann. The filter designed is shown in *Figure 5*.

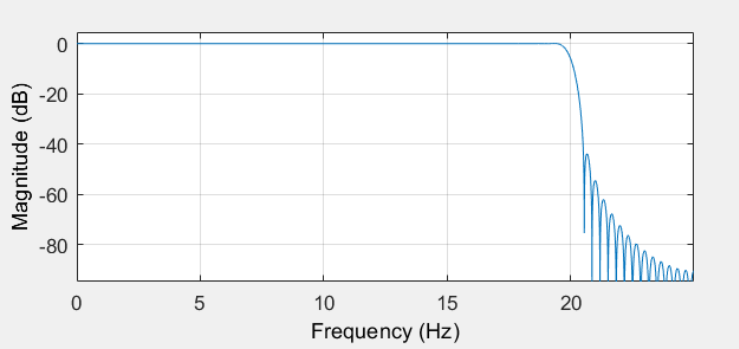


***Figure 5**: filter designing using the MATLAB application "Filter Design"*

Some advantages of FIR filters are that they can easily be implemented to have linear phase, so that the filtered signal is not distorted, and they are always stable. [3]

The method we applied to design the Filter is the Windowing method. Since our data is about vibrations, we choose the window Hann that can be used for vibration data. FIR filters have generally better performance compared to the IIRs if the number of parameters is high. The IIR filters have higher processing speed, but for this work the processing speed was not an issue.

A clear representation of the effect of the filter can be obtained plotting the Power Spectral Density (PSD) estimate obtained with the pwelch function before and after the application of the filter.

This plot shows that after the normalized frequency of 0.8, which corresponds to 20 Hz, we have a fast roll-off with a good stopband attenuation.

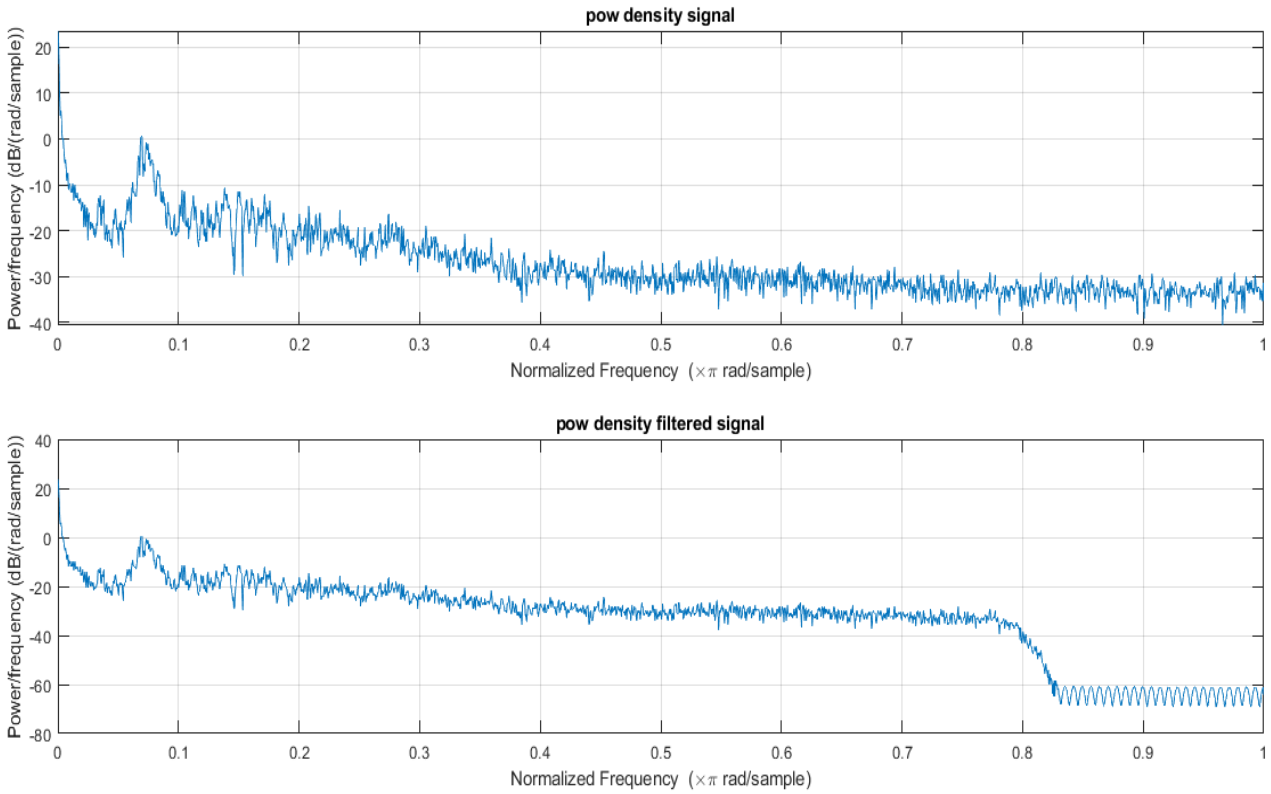


***Figure 6**: Power Spectral Density of normal and filtered signal calculated with 'pwelch' function*

5. Discrete Wavelet Decomposition

The signals are not stationary; therefore Wavelet analysis is preferable over Fourier for analysing the frequency content over time. [4]

We applied discrete wavelet transform (DWT) on the filtered signals, in particular Fast Wavelet Transform. It allows to retain a sufficient amount of information and is easy to implement.
The DWT decomposes the signal in different coefficients for distinct frequency bands. The higher the level of decomposition, the lower will be the frequencies.
To obtain the coefficients [C,L] we used the function wavedec. After decomposing the signal, we reconstructed it at the different levels using the function wrcoef. We used 5 levels of decomposition and the mother wavelet Daubechies 3 (db3) that gave us the best results.

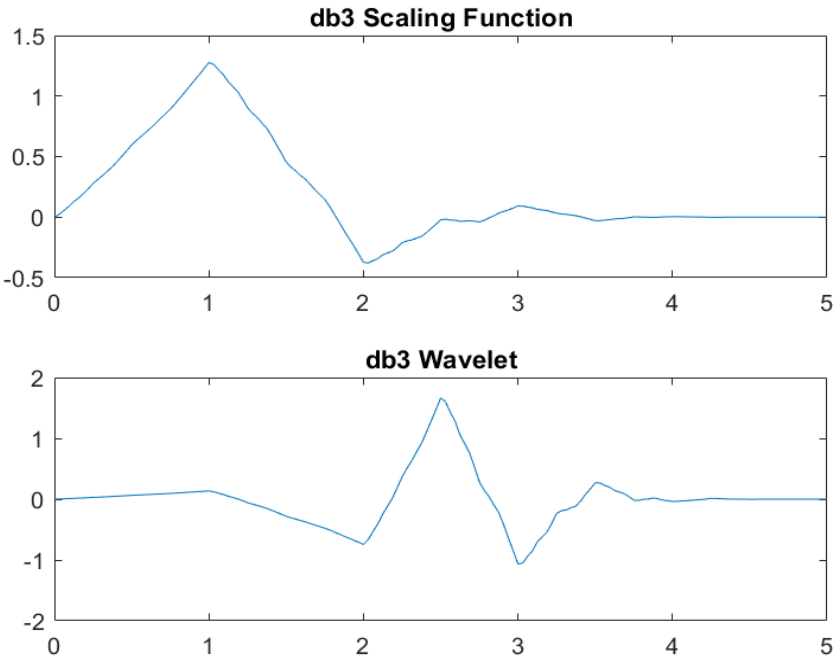


Figure 7: *Daubechies wavelet (db3)*

6. PCA & MDS

The selected features are mean, standard deviation and Shannon Entropy. All of them are extracted from each sensor and level of decomposition and correspond to different columns in the matrix. Every time-window of each experiment which is annotated with an activity is a different data point and corresponds to a row. The matrix obtained has a high number of columns, which creates dimensionality problems for clustering and makes it impossible to visualize it with a plot.

We applied dimensionality reduction methods such as PCA and MDS. Before proceeding with these methods we also reduced the number of features using Kruskal-Wallis and selecting the most relevant ones, i.e. the ones that have a higher score. We selected in total 69 features out of the 108 obtained. PCA produces a projective transformation of the data that replaces the original features with a lower number of new features obtained as their linear combination ordered by explained variance [5].

From the application of PCA we can see that only a few Principal Components are sufficient to explain almost all the variance. This justify the dimensionality reduction: we can greatly reduce the number of dimensions without losing much information: the first two principal components explain over 71% of the variance as it's shown in *figure 8*.

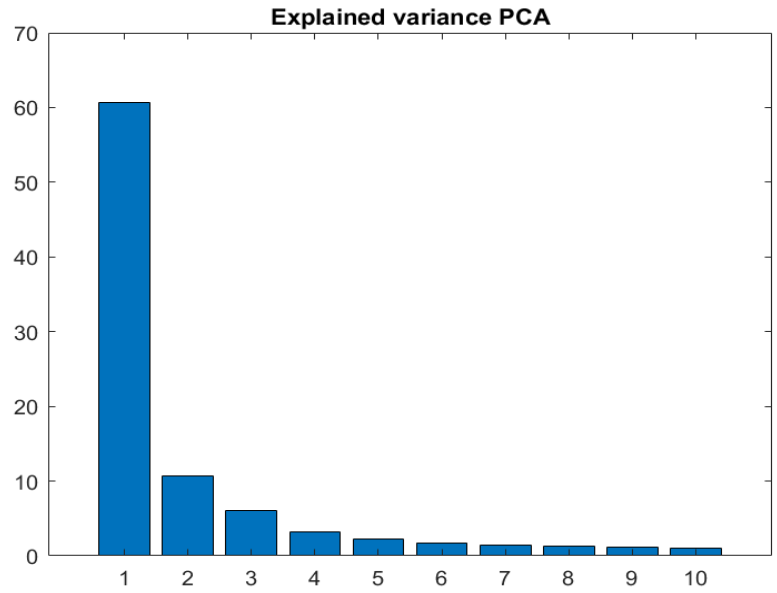


***Figure 8**: Each column represents the percentage of explained variance of a Principal Component, ordered by variance*

Multidimensional scaling is more general and allows to change the method used to calculate the distances, the criterion used to project the data and the number of dimensions.

We tried the following methods to calculate the distance: ‘euclidean’, ‘cityblock’,’cosine’ and ’correlation’.

For the mdscale function we tried the followings: ‘strain’, ‘sammon’, ‘metricstress’, ’mestricsstress’, ‘stress’,’sstress’.

We decided to use ‘euclidean’ and ‘strain’. In both cases we picked the criterion which allowed us to obtain a better separation of the data, evaluated qualitatively from the scatterplot and quantitatively in terms of accuracy of the clustering procedure, as explained in the following chapters.

The results obtained from PCA and MDS are very similar, as shown in *figure 9*. This can be explained by the fact that we are using Euclidean as the Distance criterion.

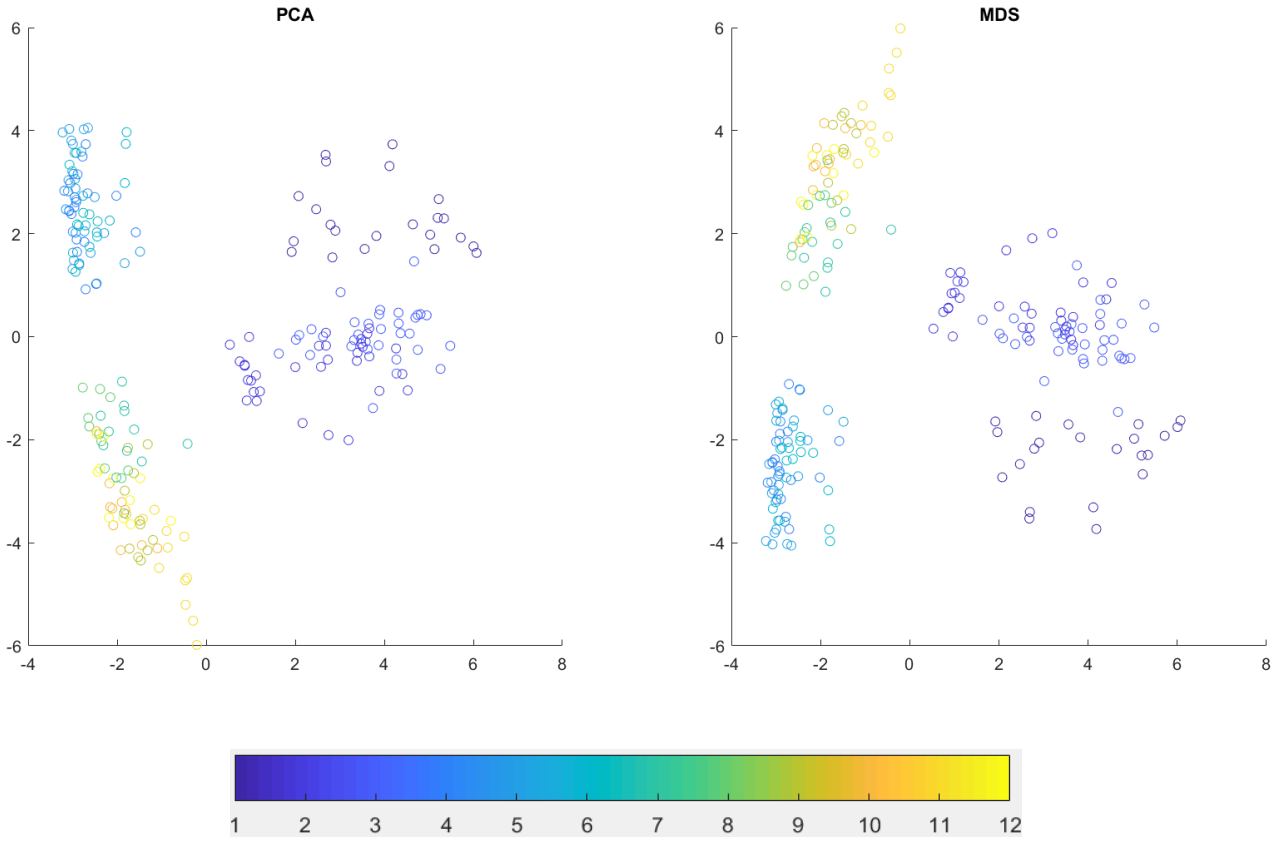


***Figure 9**: X and Y axis represent the first and second column of the Data Matrix. Each point a row. Each colour represents a different activity. On the left: Matrix reduced to 2 dimensions using PCA. On the right: Matrix reduced to 2 dimensions using MDS*

From the scatterplot we can see that the data can be separated in the three main activities: Dynamic, Static and Transition. It's not possible to separate the sub classes completely. This qualitative consideration is confirmed quantitatively in the next chapter.

7. CLUSTERING

We addressed the problem of separating the data in a known number of clusters through partition methods - k-means and k-medoids - and hierarchical clustering.

The partition methods get as input the desired number of clusters. For hierarchical clustering we specified the desired number of clusters using the parameter 'maxclust'. All the methods have been evaluated in terms of accuracy and silhouette.

Clustering for the three main classes:

As it is clear from the scatterplot in *figure10* the three main activities are completely separated so the accuracy is 100% and all the clustering algorithms give the same output, as reported in *Table* and shown in *figure 11* and *figure 12.*

	accuracy	**silhouette**
k-means	**100%**	**0.843**
k-medoids	**100%**	**0.843**
hierarchical	**100%**	**0.843**

Tab. 1: *Numerical metrics for clustering evaluation for three main activities*

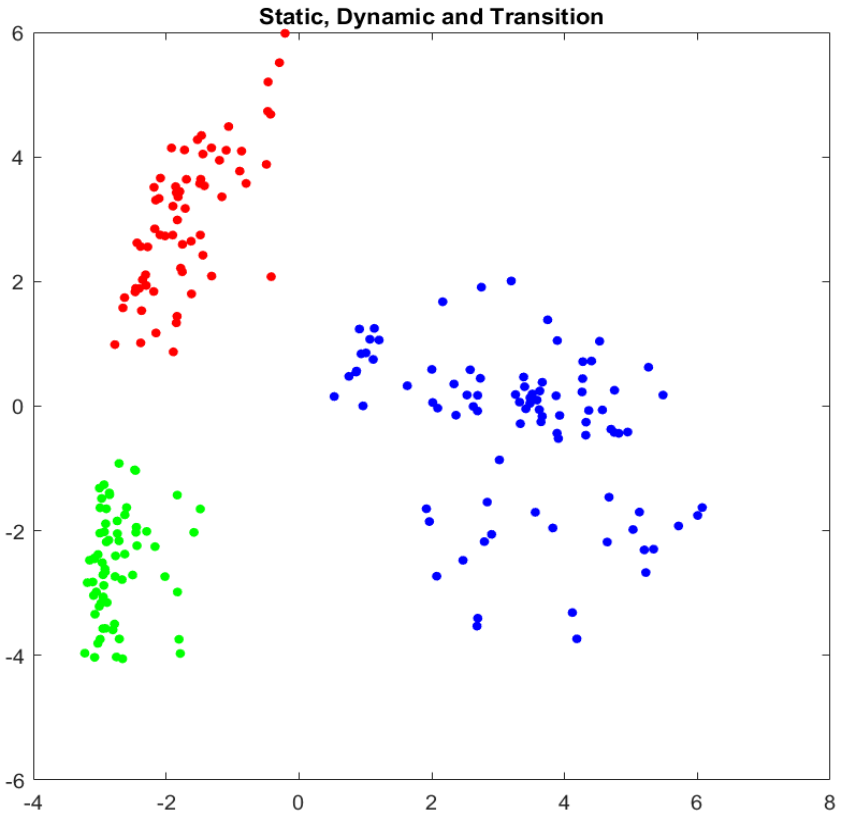


Figure 10*: Scatter plot obtained from Data Matrix reduced using MDS. Blue dots represent Dynamic activities, Green Transition and Red Static.*

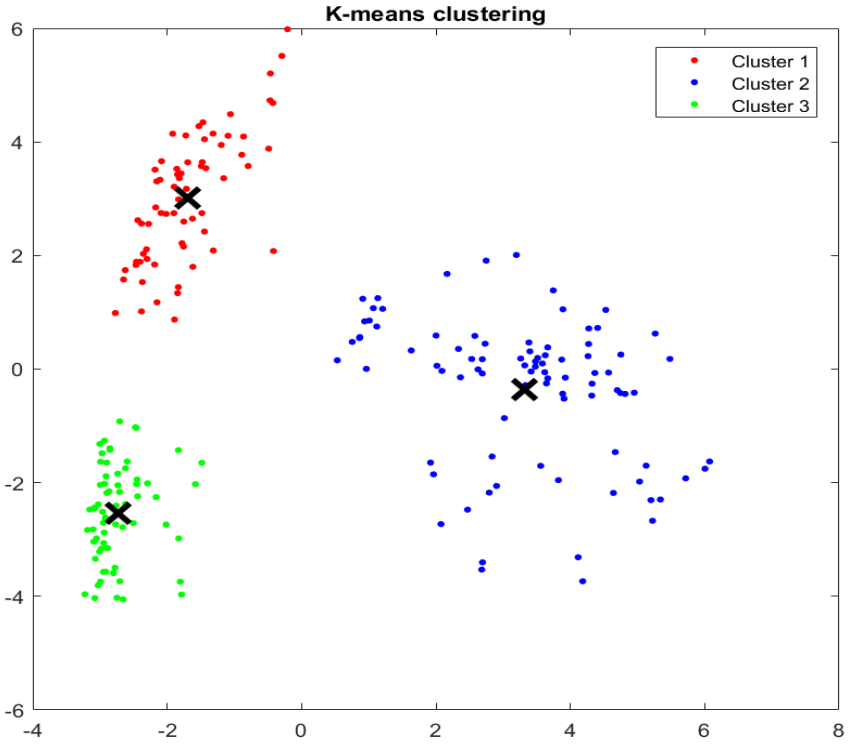


***Figure 11**: Clusters obtained applying K-means clustering. Centroids are marked as black crosses*

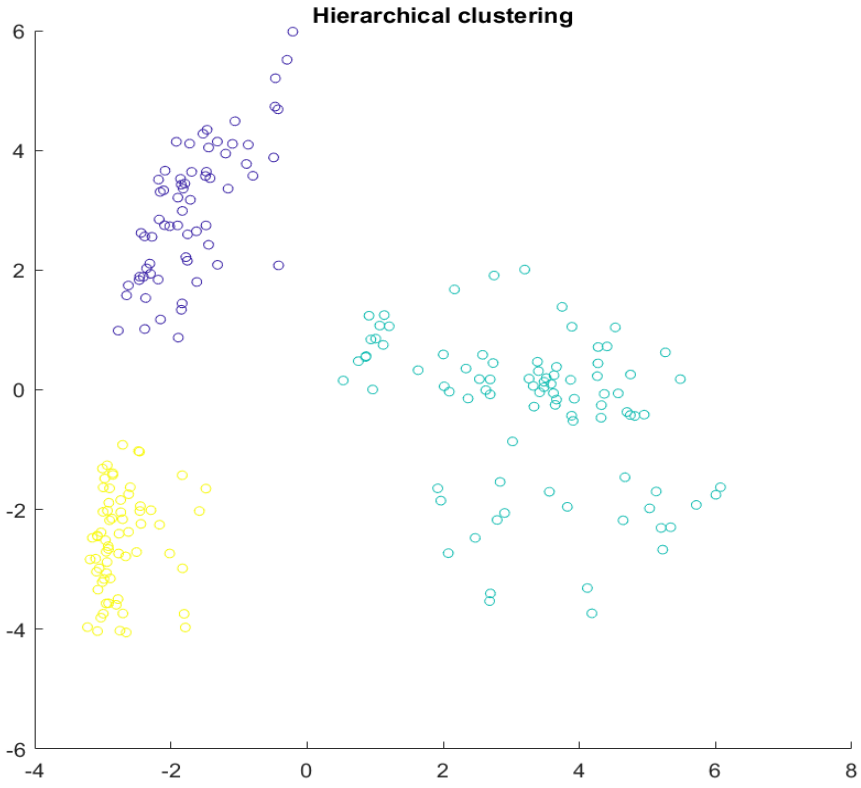


***Figure 12**: Clusters obtained applying hierarchical clustering*

Clustering for the dynamic activities:

The subclasses of the Dynamic activities (Walking, Walking up, Walking down) are not perfectly separated.

The data don't form clear clusters - as it can be seen in *figure13* - so the internal evaluation is lower than the previous case. There is some overlapping for the activities Walking up and Walking down so also the accuracy is lower. It's important to point out that we can at least correctly separate the activity *Walking* from the other two.

K-means and Hierarchical clustering produce the same result, as shown in *figure14* and *figure15*. K-medoids outputs a slightly different result which has higher cohesion but worse accuracy.

	accuracy	**silhouette**
k-means	**78.4%**	**0.616**
k-medoids	**77.3%**	**0.622**
hierarchical	**78.4%**	**0.616**

***Tab. 2**: Numerical metrics for clustering evaluation for dynamic activities*

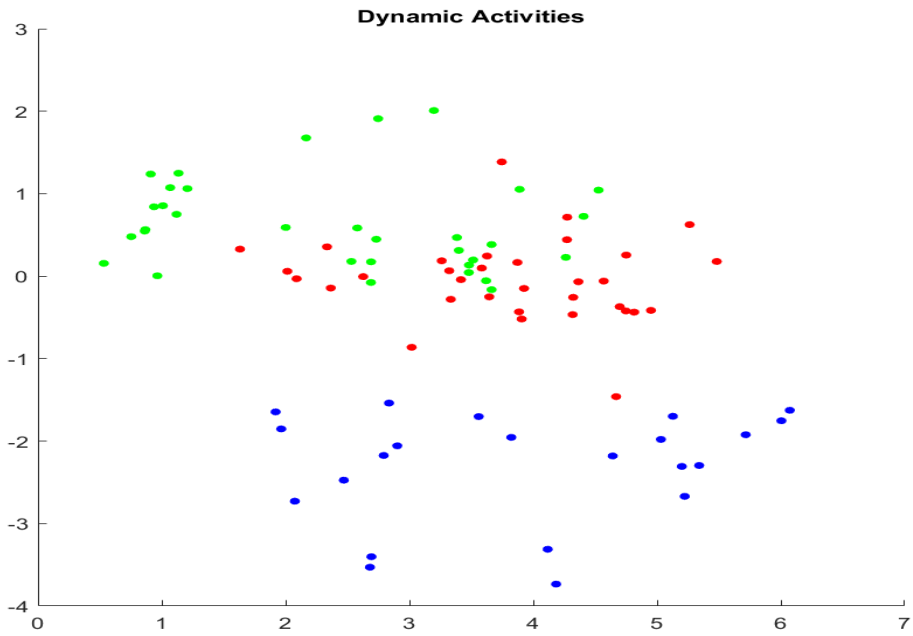


***Figure 13**: Scatter plot obtained from Data Matrix reduced with MDS. Only the rows corresponding to dynamic activities are shown. Blue: Walking. Green: Walking upstairs. Red: Walking downstairs*

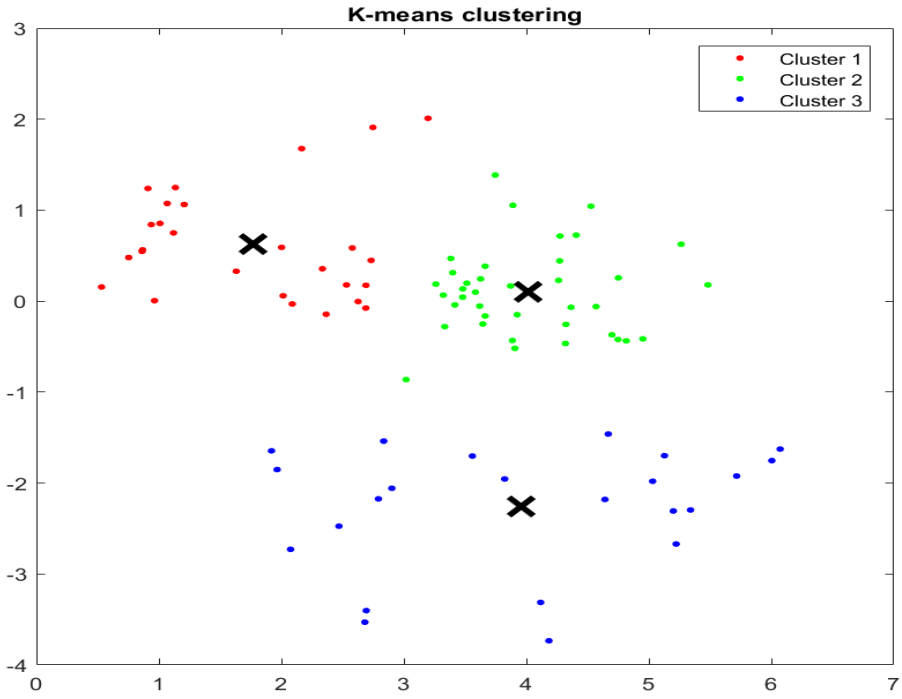


***Figure14:** Clusters obtained applying K-means clustering. Centroids are marked as black crosses*

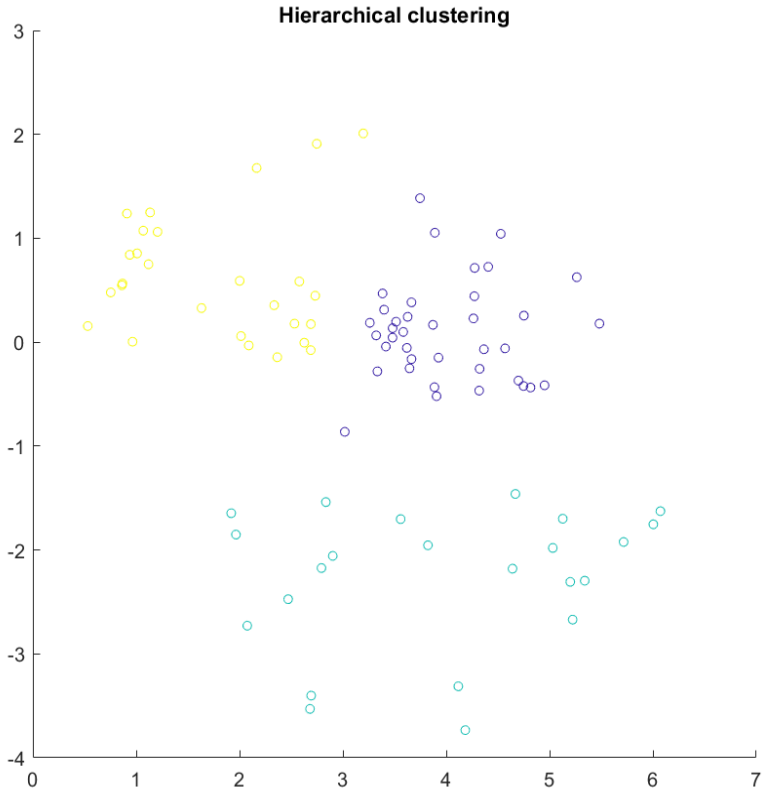


Figure 15: *Clusters obtained applying hierarchical clustering.*

Clustering for all the 12 activities:

The accuracy obtained for the 12 classes is low. The method which gives the highest score is hierarchical clustering, with an accuracy of 45%. The function implemented in MATLAB to calculate the accuracy of a clustering procedure - AccMeasure- produces an error when the number of classes is higher than 11. We decided to reduce to 11 the number of classes joining together the 11th and 12th classes. As a result, the accuracy reported could be slightly overestimated.

	accuracy	**silhouette**
k-means	**42.7**	**0.620**
k-medoids	**37.7**	**0.606**
hierarchical	**45**	**0.592**

Tab. 3: *Numerical metrics for clustering evaluation. Clustering for all 12 activities.*

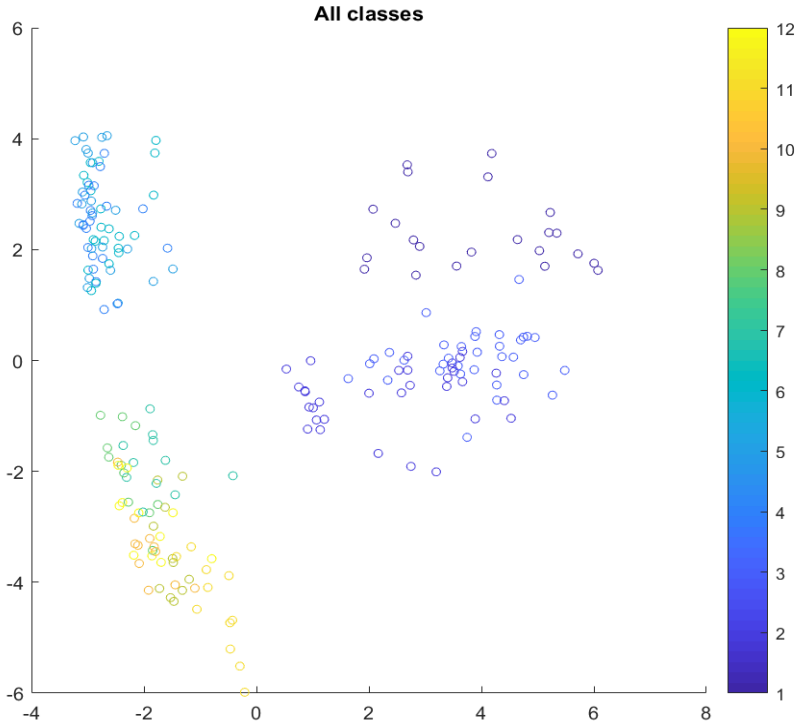


Figure 16: *Scatter plot obtained from Data Matrix reduced using MDS. Each colour represent a different activity as indicated in the colorbar*

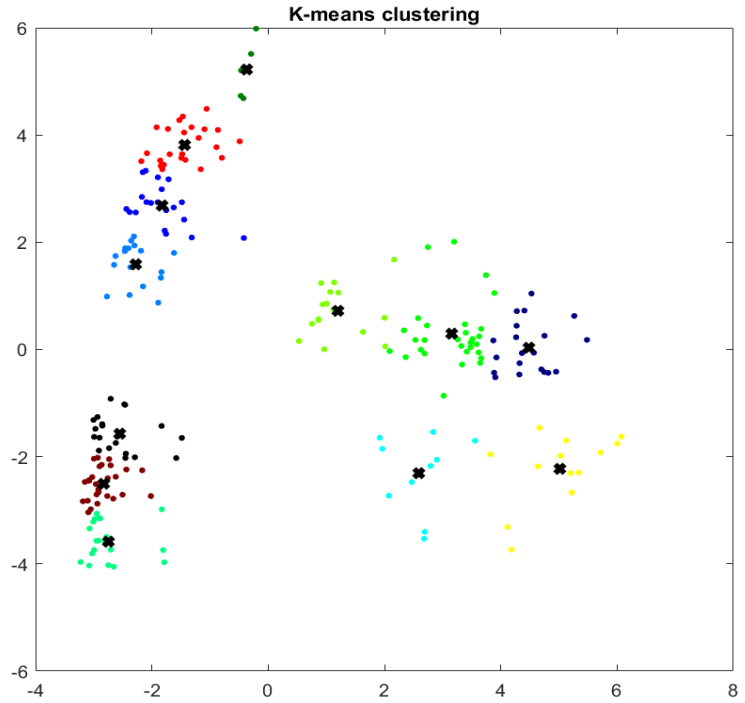


Figure 17: *Clusters obtained applying K-means clustering. Centroids are marked as black crosses*

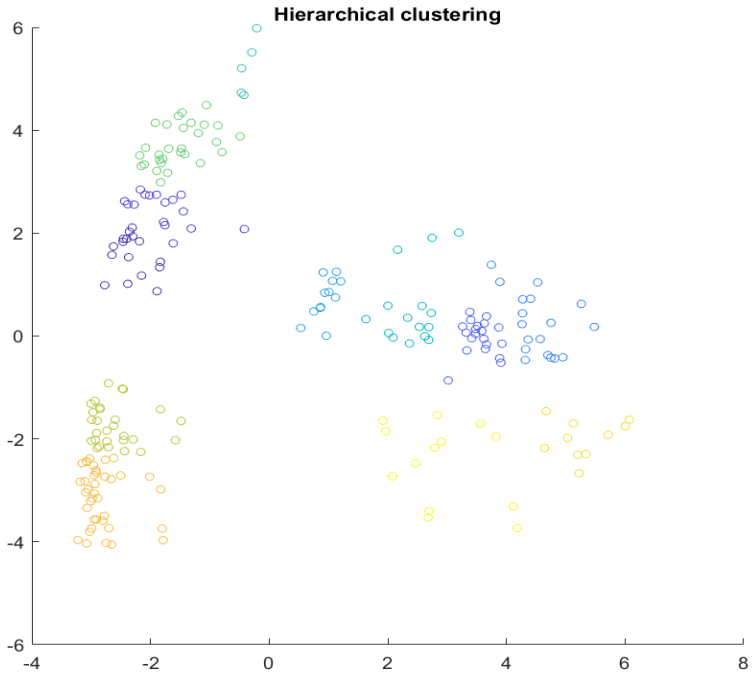


Figure 18: *Clusters obtained applying hierarchical clustering. Each colour marks a different activity.*

Silhouette evaluation:

The silhouette value is a measure of how similar an object is to its own cluster compared to the others. The range goes from −1 to +1. [6] A high value indicates that the object is well matched to its own cluster and poorly matched to neighbouring clusters. The graph shows that the silhouette values are generally slightly better for the partition methods. For all is clear that the 3 main groups - Static, Transition and Dynamic - are much better separated than the others.

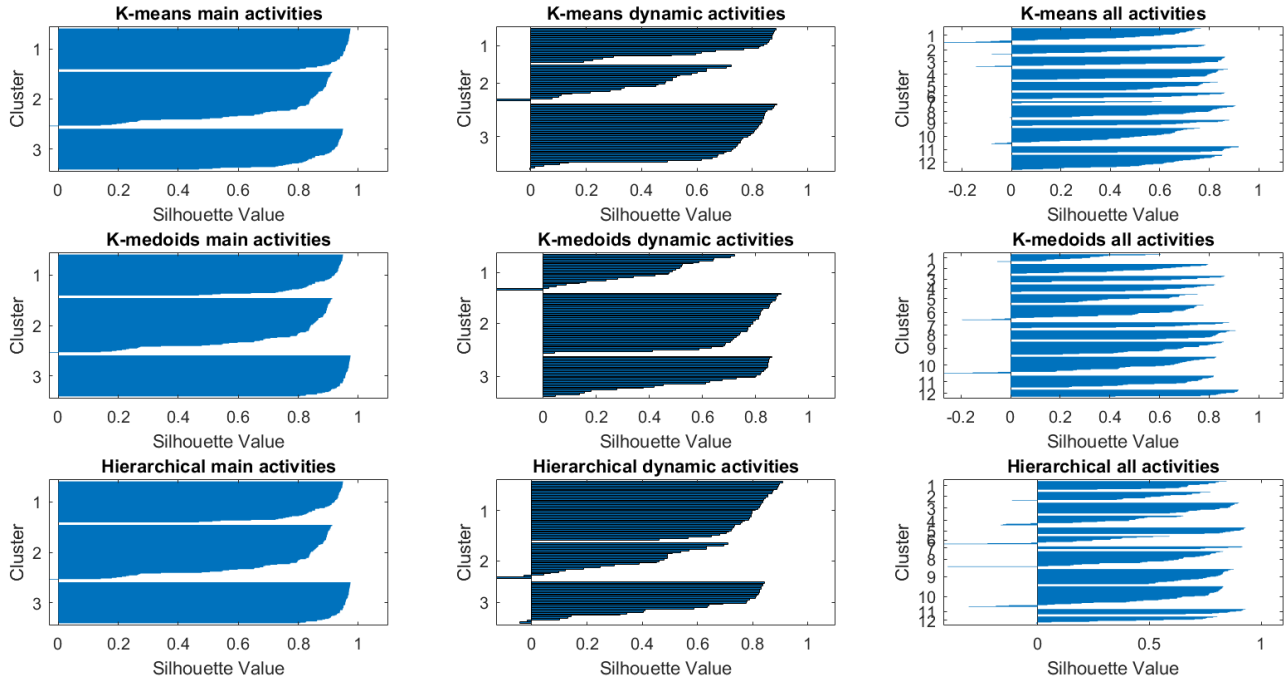


***Figure 19**: Graphical representation of the silhouette values for all the cases explained*

8. CONCLUSION

With the clustering procedure presented we were able to separate correctly the three main classes. In the separation of different dynamic activities (walking, walking upstairs and walking downstairs) the results didn't go as expected since we were able to distinguish the activity of walking but not the walking upstairs and walking downstairs. In the separation of the 12 activities the results were only partially correct.

Other methods of supervised classification not requested in this assignment allow to obtain better results. For example SVM, applied using the MATLAB application Classification Learner without reducing the dimension of the data matrix reaches accuracy of 73% for the 12 classes. The accuracy obtained with simple Clustering is only 45%.

The results could also be improved using different filtering techniques, other methods of features reduction as Correlation analysis or by extracting different features.

The data provided hold enough information to correctly classify all the activities, but the algorithms applied produced satisfactory results only for the classification of the 3 main groups.

9. References

[1] Independent Component Analysis: Algorithms and Applications. Aapo Hyvärinen and Erkki Oja. Neural Networks Research Centre Helsinki University of Technology

[2] http://research.ics.aalto.fi/ica/fastica/

[3] Signorini, 2019, Lesson 3*: Digital filtering*, lecture notes, Biomedical Signal Processing and Medical Images, Politecnico di Milano.

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[6] Silhouettes: A graphical aid to the interpretation and validation of cluster analysis. Peter J. Rousseeuw